PENG ZHAO

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LinkedIn
Google Scholar

SF Bay Area, CA

Profile

A **Camera & Optical Systems Engineer** with a proven track record of shipping solutions for multiple products from concept to mass production. Specializes in developing end-to-end simulation frameworks by integrating optics, sensors/ imaging pipelines, and virtual experiences across technical domains. Ability to synthesize design trade-offs and drive cross-functional outcomes. Experience spanning build cycles in manufacturing, testing, spec engineering for high-volume production. Holds 3 issued/filed patents and has authored over 20 peer-reviewed papers with 750+ citations.

Eye and Face Tracking Systems Engineer — 2024.2 - Present

Meta, *Sunnyvale, CA*

- Leads the optical design and architecture for advanced eye and face tracking camera systems across all Meta VR products, and roadmap explorations.
- Developed end-to-end frameworks of full optical systems including camera module, ET/FT illumination, display viewing optics, with imaging pipeline supporting high fidelity rendering. Deeply integrated with software platforms to accurately model user experience.
- Designed system level KPIs for ET/FT, established factory testing protocols, and demonstrated production quality and yield over builds.

Sensing HW Engineer — 2020.6 - 2024.3

Apple, *Cupertino, CA*

- Led eye tracking exploration for AR within laser sensing org.
- Led optical design of Apple's first ever coherent optical sensor for velocity tracking; shipped with *Apple Watch Ultra* in Fall, 2022.
- Updated optical design of an image tracking based sensor; shipped with *Apple Vision Pro* in Fall, 2023.
- Invented a new microstructure optical surface with enhanced retroflection (*patent issued*) leading to significant in cost and size reduction at system; shipped in all SKUs of Apple Watch in Fall, 2025.
- Invented a novel optical architecture of image tracking sensor that is capable of tracking both rotation and translation (*patent filed*).

Sr. Design Engineer / Team Lead — 2017.11 - 2020.6

ASML, *Wilton, CT*

- Solved polarization xtalk issue to enable the launch of first-gen polarization-resolved ORION wafer alignment sensor.
- Led technical development of a new wavelength demultiplexer module, and delivered vendor specification, proto builds, and machine integration.
- Led a cross functional team in developing a €2M production tester for next-gen product build and qualification, advancing both hardware and algorithm levels to achieve sub-nanometer motion accuracy KPIs.
- Invented a new solid-state spectral filter module for supercontinuum laser (*internal IP publication*)

Postdoc Research Associate — 2016.10 - 2017.11

CREOL, UCF, *Orlando, FL*

Funding: UCF Pre-eminent Postdoctoral Program (P^3); NSF I-Corps Site Grant

- Developed a novel nonlinear optical spectrometer based on supercontinuum.
- Developed a numerical model for double pump-probe interactions in new Ir(III)-Complexes.
- Led a NSF I-Corp program to explore commercialization of the laser refrigeration technology for thermal management in electronics.

Graduate Research Assistant — 2012.8 - 2016.10

CREOL, *UCF, Orlando, FL*

- First ever observation of nondegenerate nonlinear refraction and three-photon absorption in semiconductors and molecular responses in solvents and gases.
- Pioneer a novel two-photon, FRET-enhanced photochromic molecule for strong, broadband optical limiting application.

Graduate Research/Teaching Assistant — 2010.8 - 2012.8

Case Western Reserve University, *Cleveland, OH*

- Designed (using FDTD) and fabricated the micro-dome structures to enhance light extraction efficiency of III-nitride based LEDs.

Education

University of Central Florida, CREOL — 2012.8 - 2016.10

Ph.D. in Optics and Photonics; Advisors: Eric W. Van Stryland/David J. Hagan

Dissertation: Ultrafast Mechanisms of Nonlinear Refraction and Two-photon Photochromism

Case Western Reserve University, EECS — 2010.8 - 2012.8

M.S. in Electrical Engineering; Advisors: Hongping Zhao

Thesis: Analysis of Light Extraction Efficiency Enhancement for Deep Ultraviolet and Visible Light-Emitting Diodes with III-Nitride Micro Domes

Tianjin University, Precision Instruments Optoelectronics Engineering — 2006.8 - 2010.8

B.S. in Measuring and Controlling Technology and Instruments

Thesis: Design and optimization of mobile LED backlight module.

Patents

"Cylindrical Retroreflector Array for Rotation Tracking" - US Patent Publication No. US20230086536A1 – Date of Publication: 03/23/2023 (Issued)

"Optical Sensing of Translational and Rotational Shaft Movements" - US Patent Application No. US63541246 (filed)

""Single Photon Avalanche Diodes for Rotational and Translational Input Detection", Application No. US18892374 (filed)

Publications

13 Journals, ***8*** Refereed Proceedings, and ***28*** Conference Presentations.

Google Scholar (Link):

- Total Citations: ***750+*** (as of 2025.4); h-index: ***11***